

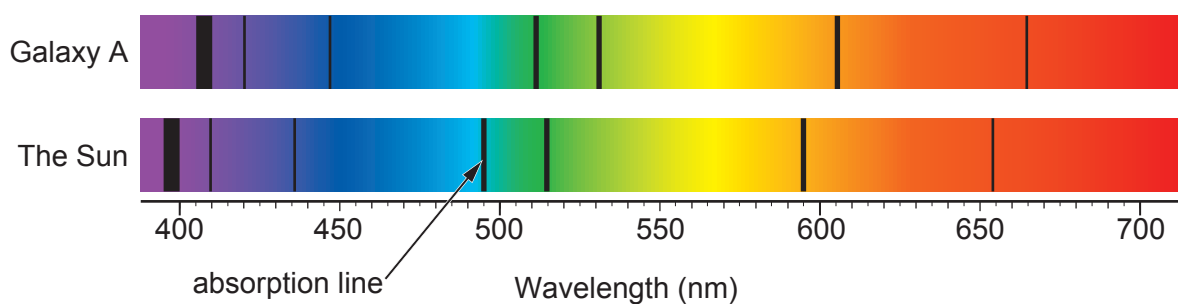
7. (a) (i) State the two pieces of evidence that support the Big Bang model of the origin of the Universe. [2]

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- (ii) The diagram below shows the absorption spectrum of a galaxy, A, compared to part of the Sun's absorption spectrum.



One absorption line in the spectrum from the Sun is labelled.
The wavelength of this line is different in galaxy A.
Explain why the wavelength of the line is different in galaxy A.

[2]

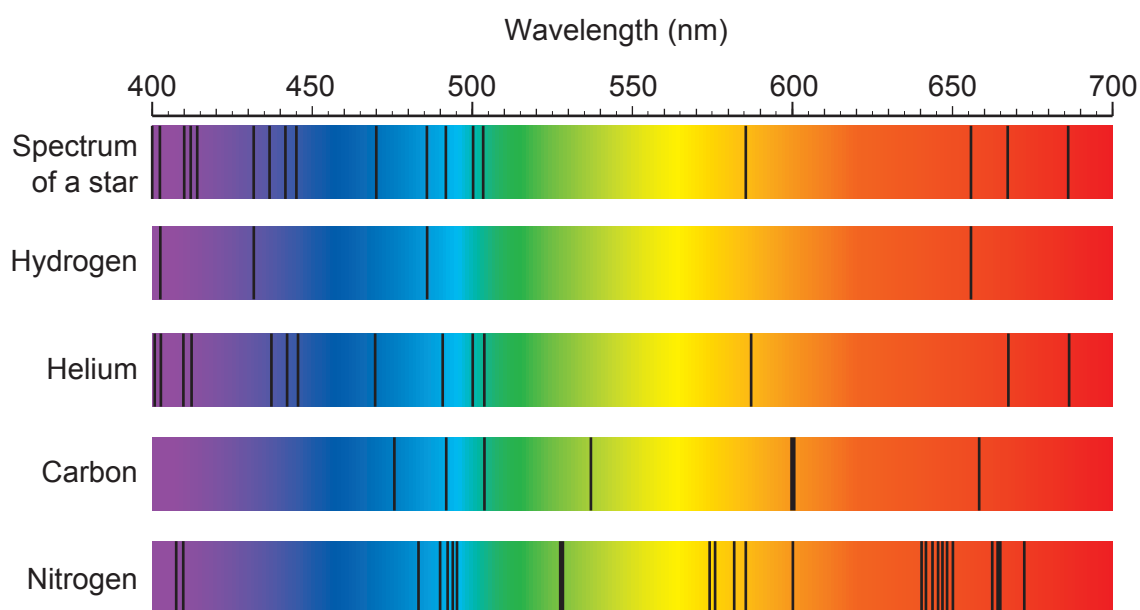
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- (b) The diagram below shows the absorption spectrum of a star and the absorption spectra of some different elements.



Matthew suggests that the star contains hydrogen and carbon.
Explain whether you agree.

[2]

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6

